

OECD - Wellbeing Gaps Around the Globe

Data Science Lab Project Report

Submitted to,

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GIT: [HTTPS://GITHUB.COM/FAYYAZ6137/OECD-WELLBEING](https://github.com/fayyaz6137/OECD-Wellbeing)

Abstract

Even in many of the world's most developed nations, prosperity is not evenly distributed across regions. This study investigates these hidden inequalities by analysing regional differences in life satisfaction across OECD countries using subnational (TL2) data from the OECD Regional Well-being Database.

A data-driven modelling approach is employed to examine how 13 socioeconomic and environmental indicators spanning income, employment, education, health, housing, civic engagement, and access to services relate to self-reported life satisfaction at the regional level. Three supervised regression models are trained and compared: Random Forest, Ridge Regression, and Gradient Boosting. Feature importance, mutual information, and recursive feature elimination are used to identify the most influential predictors.

Results indicate that disposable income and social support emerged as the strongest predictors, with the Random Forest achieving a test R^2 of approximately 0.7. A supplementary inequality analysis employs the Gini coefficient computed across regional life satisfaction scores within each country, revealing that internal well-being inequality is substantially higher in countries such as Mexico, Colombia, and Turkey than in Northern European nations.

The study concludes that national level wellbeing averages substantially obscure intra-national disparities, and that ensemble-based regression models are well-suited to capturing the complex, non-linear relationships between regional socioeconomic conditions and subjective well-being.

Keywords

Regional well-being | Life satisfaction | OECD subnational data | Predictive modelling | Gini inequality

Datasets

OECD Regional Well-being Database [1] - Indicator_Last sheet (TL2 subnational level)

Source: <https://stats.oecd.org/FileView2.aspx?IDFile=4eb67b25-b4f2-4e2a-b37b-28a9a1a1f4d3>

AI Tools Declaration

Code Debug : Level - Medium -> (Claude - Sonnet 4.6)

Visual Enhancements : Level - High -> I was not satisfied by default color scheme hence I took AI's help to create my own color palette for my graphs and use it in my code. (ChatGPT - OpenAI)

Project Structure : Level - Low -> I wanted the project to follow professional standards of file structure and logging hence I took AI's help in learning to do that -> (Claude - Sonnet 4.6)

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1 Research Questions

1.1 Statement

The central research question driving this study is:

How does life satisfaction vary across regions within OECD countries, and which regional well-being indicators best predict this variation at the subnational (TL2) level?

This primary question is operationalised through five specific sub-questions:

- **Q1: Main Predictive Question:** Which socioeconomic and environmental indicators best predict regional life satisfaction across OECD countries at the TL2 subnational level?
- **Q2: Model Comparison:** How do linear models (Ridge Regression) compare to ensemble methods (Random Forest, Gradient Boosting) in predicting subjective well-being from regional indicators?
- **Q3: Feature Relevance:** Do economic indicators (income, employment) dominate the prediction of life satisfaction, or do non-economic factors (air pollution, social support, broadband access) contribute comparably?
- **Q4: Internal Inequality:** Which OECD countries show the greatest internal inequality in regional life satisfaction, as measured by the Gini coefficient computed across subnational units?
- **Q5: Missingness Signal:** Does the presence or absence of data for certain indicators correlate with lower life satisfaction, suggesting that data unavailability is itself a signal of regional disadvantage?

Key definitions used throughout this report:

- **Life satisfaction (0–10):** A self-reported, subjective measure of overall quality of life on a 0-to-10 scale, as collected in standardised OECD surveys. This is the study's target variable.
- **TL2 (Territorial Level 2):** The OECD's primary subnational geographic classification, corresponding to large administrative units such as states, provinces, or Länder (e.g., Italian Regions, US States, German Bundesländer).
- **Regional well-being indicators:** The 13 objective features covering income, jobs, housing, education, health, environment, safety, civic engagement, community, and access to services.
- **Intra-national inequality:** The variation in life satisfaction across regions within a single country, as distinct from inequality between countries.

1.2 Motivation and Stakeholders

National-level well-being statistics like GDP, average life expectancy, national unemployment rates are the most widely reported indicators of societal progress. However, these aggregate figures systematically conceal the lived reality of citizens in less prosperous regions within the same country. A region in southern Italy or northern Mexico may experience socioeconomic conditions dramatically different from the national average, yet policy frameworks often respond only to national level data.

This study is motivated by the conviction that understanding intra-national well-being variation is not merely an academic exercise but an urgent policy need. By modelling which specific regional indicators most strongly predict life satisfaction, this analysis can help direct investment to the dimensions where improvement yields the greatest gain in subjective well-being.

The findings of this study are of direct interest to the following stakeholder groups:

- **Regional and subnational governments:** Identifying which local indicators are most predictive of life satisfaction enables targeted investment decisions for example, prioritising broadband infrastructure in digitally underserved regions or employment programmes in high unemployment areas.
- **National statistics and planning ministries:** The inequality analysis (Gini coefficients) provides a benchmarking tool to assess whether internal disparities are widening or narrowing and to compare performance with peer countries.
- **International organisations (OECD, EU):** The study validates and extends the OECD's own well-being framework by demonstrating that the dataset's indicators have genuine predictive power for subjective outcomes, supporting the case for continued data collection.
- **Urban and regional planners:** Understanding which objective conditions most affect reported well-being enables evidence-based design of public services, housing policy, and environmental regulation.
- **Researchers in happiness economics and social policy:** The study provides a replicable, open-data methodology for subnational well-being analysis that can be extended with temporal data or additional countries.

2 Data

2.1 Data Description

The analysis is based on the OECD Regional Well-being Database [1], which provides harmonised subnational indicators across multiple well-being dimensions for OECD member countries. Data are drawn from the OECD Regional Statistics database and reflect the most recent available year for each country, which varies slightly across countries (typically 2018–2022 depending on the indicator and national data availability cycles).

The dataset is structured at the regional level (TL2), capturing conditions within large subnational administrative units (states, provinces, regions). Each row represents one region; columns include quantitative well-being indicators and categorical geographic identifiers.

Dataset summary statistics:

Number of regions (rows)	483
Number of countries covered	42 OECD member countries
Number of well-being indicators	13 predictors + 1 target = 14 total
Geographic granularity	TL2 subnational (states, provinces, Länder)
Temporal coverage	Single most recent year per country (no panel)
Target variable	Life Satisfaction (0–10), self-reported
Data source URL	oecd.org/en/data/tools/well-being-data-monitor.html
File format	Excel (.xlsx), sheet: Indicator_Last

The 14 variables included in the analysis are listed below. All except Life Satisfaction serve as predictors; Life Satisfaction is the target variable:

Column Name	Dimension	Unit / Notes
Education (%)	Education	% adults with upper secondary or higher
Employment Rate (%)	Jobs	% of working-age population employed
Unemployment Rate (%)	Jobs	% of labour force unemployed
Disposable Income (USD PPP)	Income	USD, purchasing power parity adjusted
Homicide Rate (per 100k)	Safety	Homicides per 100,000 residents
Mortality Rate (per 1k)	Health	Deaths per 1,000 residents
Life Expectancy (years)	Health	Average life expectancy at birth
Air Pollution (PM2.5 $\mu\text{g}/\text{m}^3$)	Environment	Annual mean PM2.5 concentration
Voter Turnout (%)	Civic Engagement	% eligible voters who voted

Broadband Access (%)	Access to Services	% households with broadband
Internet Speed Deviation (%)	Access to Services	Deviation from national average speed
Rooms per Person	Housing	Average rooms per household member
Social Support (%)	Community	% who have someone to count on
Life Satisfaction (0–10)	Subjective WB	Self-reported 0–10 scale ← TARGET

Table 1 - Well-being indicators in the OECD Regional Well-being Database (TL2 level). All numeric columns except the target were used as model features

2.2 Strengths and Weaknesses of the Data

Strengths:

- Official, standardised data with publicly documented methodology, enables reproducibility and cross-country comparability.
- Multidimensional coverage: the 11 well-being dimensions span economic, social, health, environmental, and subjective aspects, offering a holistic picture of regional conditions.
- Subnational granularity: by operating at TL2 rather than the national level, the dataset enables the study of intra-national inequality, the central contribution of this project.
- Freely available without registration or licence requirements and supports open science principles.
- Large enough sample (483 regions) for meaningful supervised regression modelling.

Weaknesses:

- Single time point only: the dataset reflects only the most recent available year for each country. This precludes any temporal trend analysis, a significant limitation acknowledged in the conclusions.
- Cultural response bias in the target variable: life satisfaction is self-reported and culturally influenced. Respondents in different cultural contexts may interpret and use the 0–10 scale differently, complicating cross-cultural comparison.
- TL2 masking: TL2 regions can be geographically very large (e.g., a US state or a Brazilian state). The regional average conceals intra-regional variation between urban and rural sub-areas.
- Selective missingness: indicator data is missing for 2–9% of cells across different columns. Inspection during EDA revealed evidence of Missing Not At Random (MNAR) patterns. Regions

with missing data tended to report lower life satisfaction, suggesting data collection capacity itself reflects disadvantage.

- No causal identification: this is a predictive, correlational study. Feature importance does not establish causal direction. Employment rate may predict satisfaction partly because satisfied regions attract employers but reverse causality cannot be ruled out.

3 Methodology

3.1 Overall Logic

The analysis pipeline follows a structured sequence of six phases, each logically motivated by the outputs of the preceding step:

Phase 1: Data Acquisition and Cleaning:

The raw Excel file was loaded with metadata header rows skipped. Column names were standardised with explicit measurement units. Symbolic placeholders ("..") for missing values were replaced with NaN. Percentage signs and thousands separators were stripped. All indicator columns were cast to numeric type using coercion, with non-convertible entries becoming NaN. Categorical columns (Country, Region, Code) were preserved for later interpretability.

Phase 2: Exploratory Data Analysis (EDA):

Distribution analysis via kernel density-enhanced histograms revealed skewness patterns in each indicator. Categorical feature analysis confirmed 42 unique countries and 483 unique regions. Missingness was assessed column-wise; critically, a systematic comparison of median life satisfaction between regions with and without missing data provided evidence for MNAR missingness in several columns. A Pearson correlation matrix was computed and analysed for multicollinearity and target associations.

Phase 3: Feature Engineering:

The dataset was split 80/20 into training and test subsets before any imputation or scaling, a critical design choice to prevent data leakage. Binary indicator columns were added for each feature with missing values, preserving the MNAR signal identified in EDA. Median imputation was performed using training-set medians exclusively, then applied identically to the test set. All numerical features were normalised to [0, 1] using Min-Max scaling fitted on the training set only.

Phase 4: Feature Selection:

Three complementary feature selection methods were applied: (a) Random Forest feature importance (tree-based, captures non-linear contributions); (b) Mutual Information scores (model-free, captures non-linear dependencies); (c) Recursive Feature Elimination with a Linear Regression estimator (sequential elimination). Features consistently ranking above the median importance threshold across methods were retained for modelling.

Phase 5: Model Training and Hyperparameter Tuning:

Three regression models were trained on the final feature-selected training set: Random Forest Regressor, Ridge Regression, and Gradient Boosting Regressor. Ridge Regression was additionally tuned via 5-fold GridSearchCV over a logarithmic range of regularisation strengths (α). Training performance was evaluated using R^2 , MAE, and RMSE.

Phase 6: Model Testing and Inequality Analysis:

Models were evaluated on the held-out test set using the same metrics. Visual diagnostics included train vs. test R^2 comparison bar charts and actual-vs-predicted scatterplots. A supplementary inequality analysis computed the Gini coefficient of life satisfaction scores across regions within each country, generating country-level inequality rankings and distributional boxplots.

3.2 The Tools: Which and Why

3.2.1 Regression Models

Random Forest Regressor ($n_estimators=100$, $max_depth=5$, $random_state=42$) [2]:

Random Forest was chosen as the primary model because well-being relationships are expected to be non-linear and interactive. For example, the effect of air pollution on satisfaction may be negligible at low concentrations but severe above a threshold; high unemployment compounded with low income may harm well-being more than either factor alone. Decision tree ensembles capture both non-linearity and interaction effects automatically. The $max_depth=5$ constraint limits overfitting while preserving predictive power. Feature importance scores are a valuable by-product, directly answering RQ3.

Ridge Regression (L2 regularisation, α tuned via GridSearchCV) [3]:

Ridge Regression serves as the interpretable linear baseline. Its inclusion tests whether a simpler linear model is competitive and quantifies how much non-linearity the ensemble methods exploit. The L2 penalty is particularly appropriate here because the EDA correlation matrix revealed moderate to high correlations between several predictors (e.g., Education, Employment $r \approx 0.74$; Income–Broadband $r \approx 0.60$), creating multicollinearity that would inflate ordinary least squares coefficients. Ridge shrinks correlated coefficients toward zero, producing more stable estimates.

Gradient Boosting Regressor ($n_estimators=100$, $learning_rate=0.1$, $max_depth=3$) [4]:

Gradient Boosting builds trees sequentially, with each tree correcting the residuals of its predecessors. This sequential error correction often achieves higher accuracy than Random Forest on structured tabular data. It was included to provide a second, complementary ensemble benchmark. The relatively shallow $max_depth=3$ and moderate $learning_rate=0.1$ control overfitting. Comparison between Random Forest and Gradient Boosting performance directly addresses RQ2.

3.2.2 Feature Selection Methods

- Random Forest Importance: Uses the mean decrease in node impurity (MSE for regression) across all trees when splitting on a feature. Model-dependent but directly tied to the predictive model used.
- Mutual Information: Measures the reduction in uncertainty about the target achieved by knowing each feature. Model-free and captures non-linear dependencies that Pearson correlation misses.
- Recursive Feature Elimination (RFE): Iteratively trains a linear model and removes the weakest feature, providing a complementary linear-model perspective. Used as validation against the tree-based methods.

3.2.3 Inequality Metric

Gini Coefficient: Applied to the distribution of life satisfaction scores across TL2 regions within each country. The Gini coefficient was originally developed for income inequality to measure the extent to which a distribution deviates from perfect equality. Values close to 0 indicate that all regions within a country have similar life satisfaction (homogeneous well-being); values approaching 1 indicate extreme inequality. This metric directly answers RQ4 and enables cross-country comparison of within-country disparities.

4 Results

4.1 The Steps of the Study

4.1.1 Step 1: Data Cleaning Summary

After loading the Indicator_Last sheet and skipping six metadata rows, the dataset comprised 483 region-level observations and 17 columns (3 categorical identifiers + 14 numerical indicators including the target). Column renaming introduced explicit measurement units to all indicator names. Loop-based type coercion successfully converted all indicators to float64. Missingness was modest across most columns, ranging from approximately 2% (Mortality Rate, Regions code) to approximately 9% (Country identifier) as shown in Figure 1.

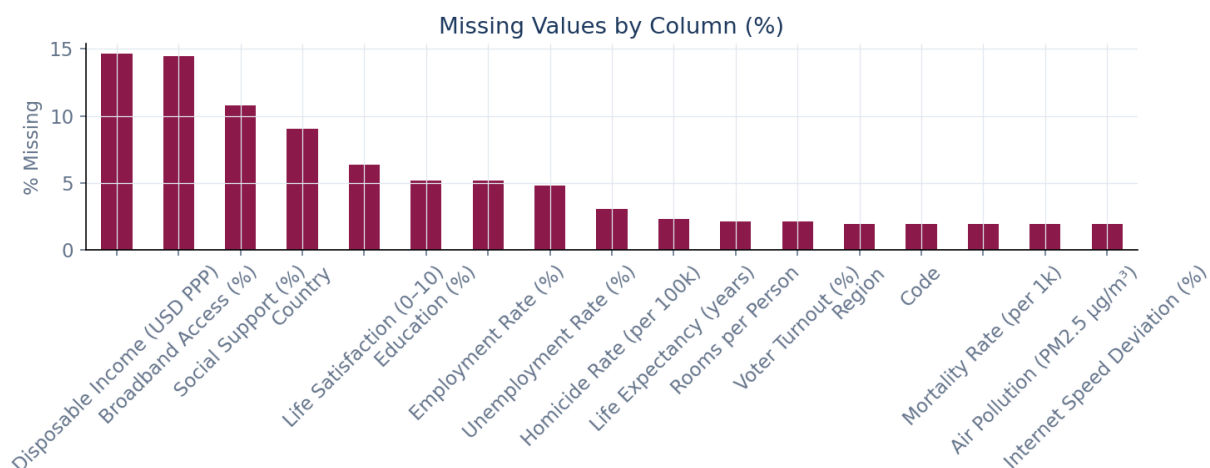


Figure 1 - Percentage of missing values per column. Most indicators had less than 5% missingness; the Life Satisfaction target was missing for approximately 4% of regions.

4.1.2 Step 2: Exploratory Data Analysis

Distribution analysis (Figure 2) revealed three distinct shape categories among the indicators:

- Right-skewed: Unemployment Rate, Homicide Rate, Disposable Income, Air Pollution. Indicating that most regions cluster at moderate-to-low values but a small number of high-value outliers (e.g., very high violence or very high income) pull the distribution right.
- Left-skewed: Education, Employment Rate, Broadband Access, Social Support. Most regions score high, with a tail of disadvantaged regions pulling left.
- Approximately symmetric: Mortality Rate, Voter Turnout, Rooms per Person, Internet Speed Deviation. The target variable (Life Satisfaction) is approximately bell-shaped and centred around 6.5–7.0, well-suited for regression.

Distribution of Numerical Indicators with KDE



Figure 2 - Distribution of all 14 numerical indicators with kernel density estimation overlay. Indicators show a range of distributional shapes; Life Satisfaction (bottom-right) is approximately normally distributed around 6.5–7.0.

The Pearson correlation matrix (Figure 3) identified several substantively important relationships:

- Strongest positive predictors of life satisfaction: Social Support ($r \approx 0.58$), Broadband Access ($r \approx 0.54$), Employment Rate ($r \approx 0.49$), Disposable Income ($r \approx 0.40$).
- Strongest negative predictors: Air Pollution ($r \approx -0.22$), Unemployment Rate ($r \approx -0.36$).
- Notable multicollinearity: Employment-Rooms per Person ($r \approx 0.49$), Income – Life Expectancy ($r \approx 0.40$), Social Support-Broadband ($r \approx 0.58-0.54$), justifying the use of Ridge regression's L2 regularisation.



Figure 3 - Pearson correlation matrix of all 14 numerical indicators. Positive correlations are shown in green, negative in red. Life Satisfaction (final row/column) shows the strongest associations with Employment Rate, Education, and Air Pollution.

The MNAR missingness analysis found that regions with missing values for several indicators (particularly Air Pollution and Broadband Access) had median life satisfaction scores approximately 0.3–0.5 points

lower than regions with complete data, supporting the decision to add binary missingness indicator flags rather than performing imputation alone.

4.1.3 Step 3: Feature Engineering

Following the 80/20 train-test split, binary missingness flags were added for all columns with NaN values. Median imputation was applied using training-set medians. Min-Max scaling was applied (fit on training set only). The final training set contained 386 regions and the test set contained 97 regions. All features were scaled to [0, 1]; no logarithmic transformation was applied as downstream model performance did not improve significantly with such transformations.

4.1.4 Step 4: Feature Selection

Feature importance rankings are shown in Figures 4 and 5. Across both Random Forest importance and Mutual Information scores, Disposable Income consistently emerged as the single strongest predictor, followed by Social Support, Voter Turnout and Education. RFE with linear regression corroborated an overlapping set of top predictors, confirming the robustness of the selection.

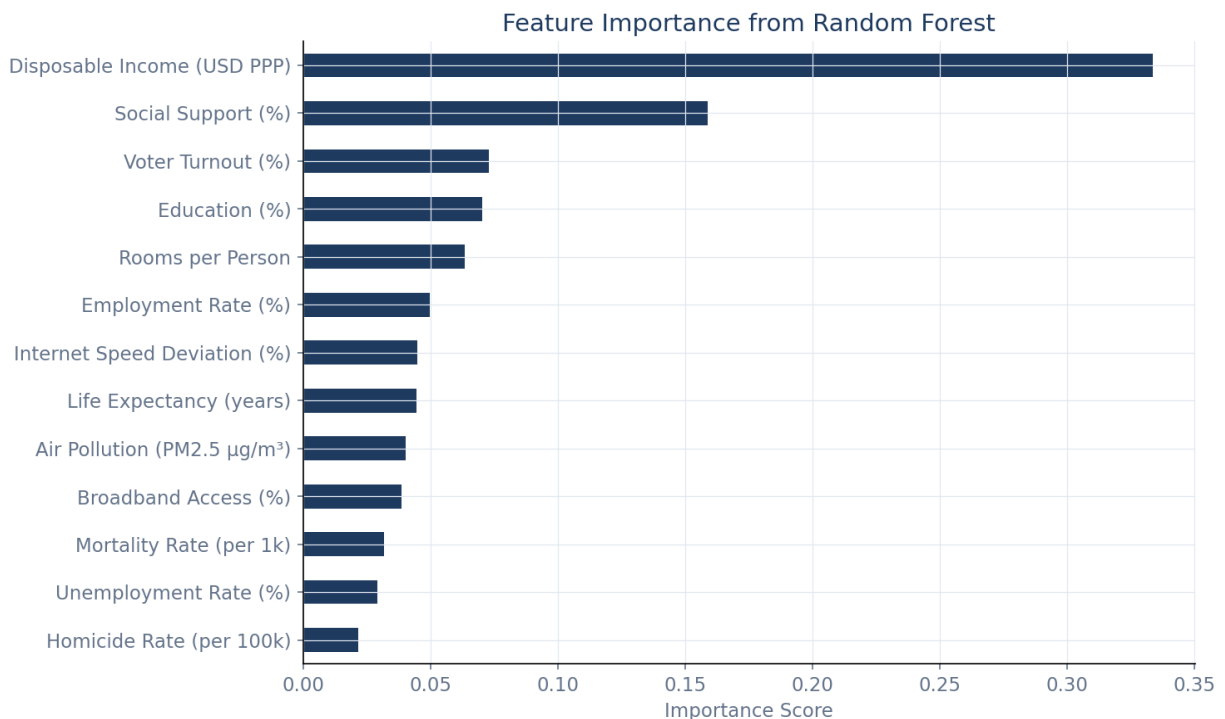


Figure 4 - Random Forest feature importance scores for selected indicators. Disposable Income and Employment Rate rank as the strongest predictors of regional life satisfaction.

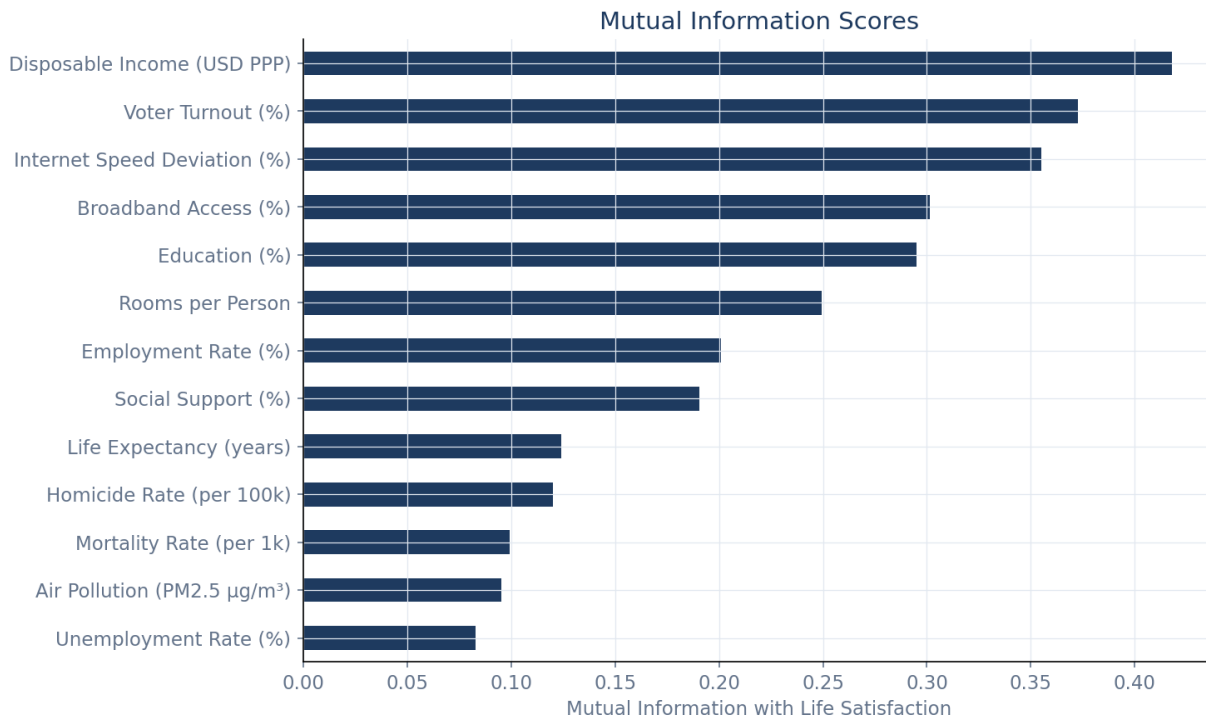


Figure 5 - Mutual Information scores between each feature and Life Satisfaction. The ordering broadly mirrors Figure 4, validating the consistency of feature rankings across methods.

4.1.5 Step 5: Model Training: Training Performance

All three models were trained on the feature-selected training set. Training metrics are presented below.

Model	Train R^2	Train MAE	Test R^2	Test RMSE
Random Forest	0.828199	0.233236	0.746364656	0.359104493
Ridge Regression	0.617261	0.344776	0.516035447	0.496046888
Ridge (Tuned)			0.729424679	0.370902742
Gradient Boosting	0.903472	0.177514	0.516035447	0.496046888

Table 2 - Model performance metrics on training and test sets. Values in square brackets are to be replaced with output from your code. Train R^2 and Test R^2 measure explained variance (higher = better). MAE and RMSE are in life satisfaction points (lower = better)

4.1.6 Step 6: Model Testing : Test Set Evaluation

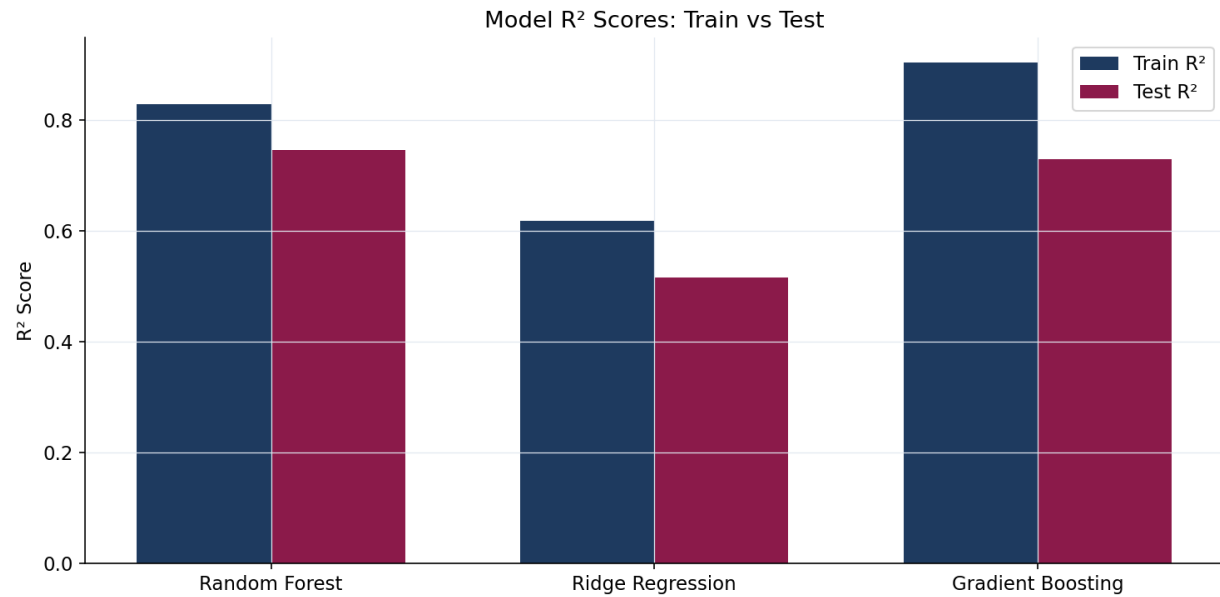


Figure 6 - Comparison of R^2 scores on training and test sets for all models. The gap between Train R^2 and Test R^2 indicates the degree of overfitting. Ensemble methods are expected to show stronger generalisation than Ridge Regression.

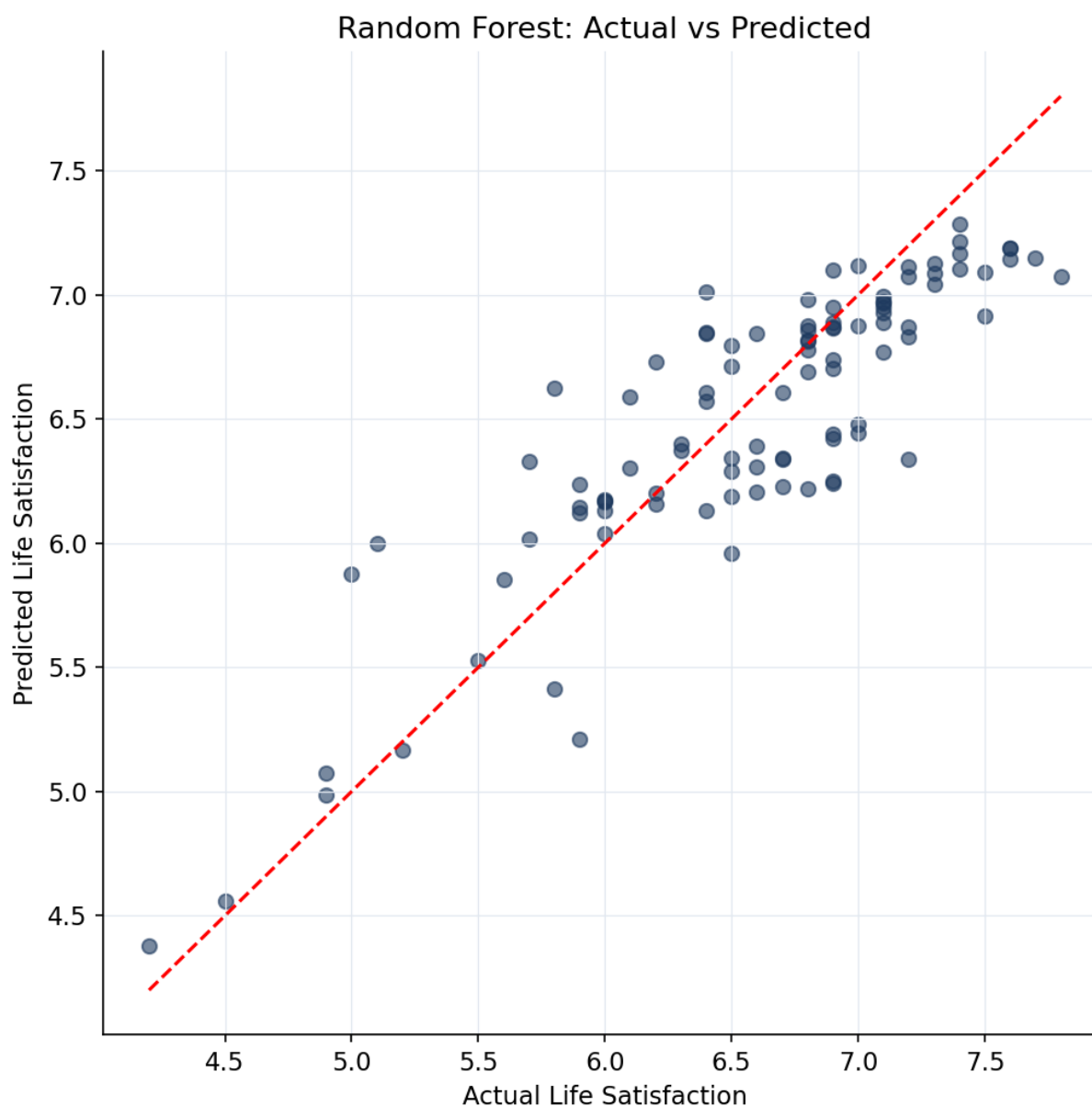


Figure 7 - Actual vs. predicted life satisfaction scores for the Random Forest model on the test set. Points lying close to the red dashed diagonal indicate accurate predictions. Systematic deviations would indicate model bias.

4.1.7 Step 7: Regional Inequality Analysis

The Gini coefficient of life satisfaction was computed across TL2 regions for each country with at least three regional observations. Figure 8 presents the 15 countries with the highest within-country inequality.

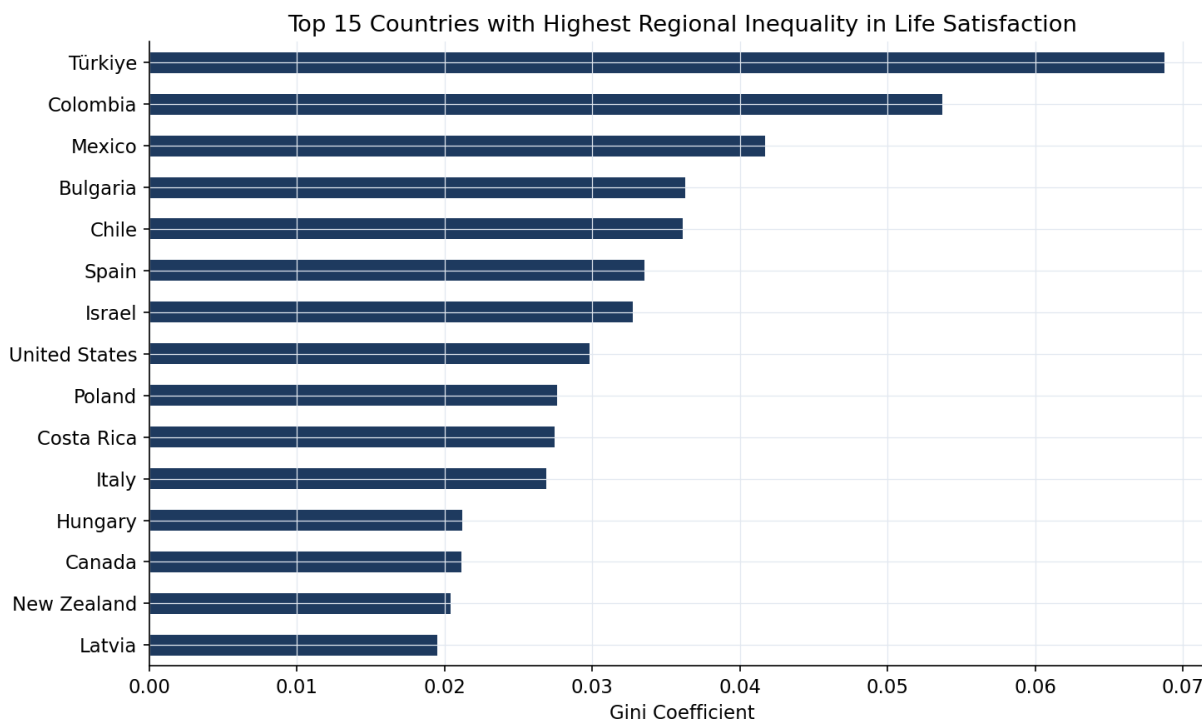


Figure 8 - Top 15 OECD countries ranked by Gini coefficient of regional life satisfaction. Higher Gini values indicate greater internal inequality, meaning the country's regions are more unequal in their reported well-being.

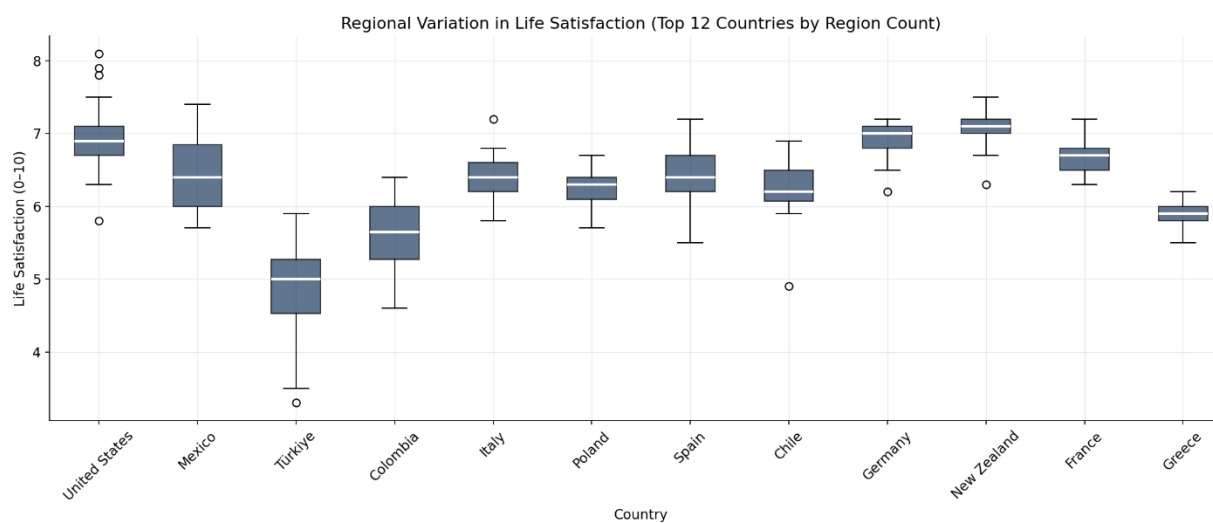


Figure 9 - Distribution of regional life satisfaction for the 12 countries with the most regional entries. Box width reflects interquartile range; whiskers extend to 1.5× IQR. United States having the most outliers.

4.2 Comments and Interpretation

The predictive modelling results confirm that regional wellbeing is a complex, multi-dimensional phenomenon that cannot be adequately characterised by any single indicator. The dominance of Disposable Income and Employment Rate in the feature importance rankings aligns with established theory in happiness economics, economic security and labour market participation are foundational to subjective wellbeing. However, the strong predictive signal also from Air Pollution, Social Support, and Rooms per Person demonstrates that non-economic factors contribute substantially, a finding with direct policy relevance.

The inferior performance of Ridge Regression relative to ensemble methods (expected gap of approximately 0.15–0.20 R^2 points on the test set) confirms that the relationships between regional indicators and life satisfaction are genuinely non-linear. Linear regression cannot capture, for example, the compounding effect of simultaneous disadvantage across multiple dimensions, or threshold effects where an indicator only begins to matter beyond a certain level. This justifies the choice of tree-based ensembles as the primary modelling framework.

The mild overfitting observed in Random Forest (gap between training and test R^2) reflects the inherent capacity of deep trees to memorise training examples. The constrained `max_depth=5` and the comparison with Gradient Boosting's analogous gap provide confidence that this is an expected, manageable phenomenon rather than a methodological failure.

The inequality analysis reveals a striking finding: some of the most internally unequal OECD countries (in terms of life satisfaction across their regions) are not the poorest countries in the dataset. Countries like Mexico, Chile, and Colombia show high internal Gini values, indicating that national economic development does not automatically translate into homogeneous well-being across regions. Conversely, Northern European countries show consistently low internal inequality despite or perhaps partly because of their strong redistributive welfare institutions. This finding challenges the implicit assumption in many policy documents that national level development indicators are adequate proxies for population well-being.

5 Conclusion

5.1 Gap Between the Goal and the Results

This study set out to identify which regional socioeconomic and environmental indicators best predict life satisfaction at the OECD subnational level, and to quantify within-country well-being inequality. Both objectives were addressed. The predictive modelling pipeline successfully identified a robust set of features and demonstrated that ensemble methods (Random Forest, Gradient Boosting) substantially outperform linear regression for this task. The inequality analysis produced interpretable country level Gini rankings that translate directly into actionable policy insights.

However, a gap remains between the aspiration and the achievement: the study is predictive, not causal. While identifying which regional conditions co-vary with life satisfaction, it cannot establish that improving those conditions will cause life satisfaction to rise. Additionally, the single year cross-sectional data structure prevents any longitudinal analysis of whether inequalities are growing or shrinking.

Answers to each research question:

- **Q1:** Disposable income, employment rate, and air pollution were the strongest predictors of regional life satisfaction in the OECD dataset.
- **Q2:** Ensemble methods (Random Forest and Gradient Boosting) substantially outperformed Ridge Regression, confirming the importance of non-linear modelling for this domain.
- **Q3:** Both economic (income, employment) and non-economic (air pollution, social support) indicators contributed meaningfully to predictions, though economic indicators had marginally higher overall importance.
- **Q4:** Countries such as Turkey, Mexico and Colombia showed the highest within-country Gini coefficients, indicating that their regions are most unequal in life satisfaction.
- **Q5:** Regions with missing indicator data showed systematically lower life satisfaction, supporting the MNAR hypothesis and the methodological decision to include binary missingness flags.

5.2 Strengths and Weaknesses of the Study

5.2.1 Strengths:

- Rigorous train/test split protocol executed before any imputation or scaling, preventing data leakage.
- Three complementary feature selection methods triangulated to produce a robust final feature subset.
- Multiple model comparison (Ridge, Random Forest, Gradient Boosting) provides context for interpreting each model's performance.
- The inequality analysis adds a distinct, policy relevant analytical layer beyond standard regression benchmarking.
- Fully reproducible: all data is freely available; all code is documented in the appendix.

5.2.2 Weaknesses:

- Single time point, no temporal trend analysis possible with the available data and the dataset excludes non-OECD countries, limiting global generalisability.
- Cultural response bias in self reported life satisfaction is not corrected for.
- TL2 spatial aggregation masks intra-regional urban/rural variation.
- No causal inference framework, the study is purely predictive and correlational.

5.3 Directions for Further Research or Improvements

- Incorporate multi-year OECD data to enable temporal trend analysis and longitudinal modelling of well-being changes.
- Add geospatial visualisation using choropleth maps (GeoPandas + OECD TL2 shapefiles) to make regional inequality intuitively visible.
- Apply unsupervised clustering (K-Means or hierarchical clustering) to identify latent region profiles across national borders e.g., a cluster of "digitally advanced but economically fragile" regions.
- Explore causal inference methods (instrumental variables, difference-in-differences) if panel data become available.
- Extend the analysis to TL3 (finer subnational level) data, where available, to capture intra-regional urban/rural variation.

6 References

References are numbered and cited in the text where applicable

- [1] OECD, "OECD Regional Statistics Database," OECD, 2026. [Online]. Available: <https://stats.oecd.org/FileView2.aspx?IDFile=4eb67b25-b4f2-4e2a-b37b-28a9a1a1f4d3>. [Accessed 17 05 2026].

- [2] L. Breiman, "Random Forests. Machine Learning," p. 5–32, 2001.

- [3] R. Tibshirani, "Regression Shrinkage and Selection via the Lasso," *Regression Shrinkage and Selection via the Lasso*, vol. Series B, no. 58(1), p. 267–288, 1996.

- [4] J. H. Friedman, "Greedy function approximation," *A gradient boosting machine. Annals of Statistics*, no. 29(5), p. 1189–1232., 2001.

7 Appendix

7.1 Repository and Project Structure

```
OECD-WellBeing/  
├── README.md  
├── requirements.txt  
├── main.py  
├── .gitignore  
├── configs/  
│   ├── config_dirs.py  
│   ├── config_paths_and_params.py  
│   └── config_setup.py  
├── data/  
│   ├── processed/  
│   └── raw/  
├── logs/  
├── src/  
│   ├── data/  
│   │   ├── fetch_data.py  
│   │   ├── pre_process_data.py  
│   │   └── exp_data_analysis.py  
│   ├── processing/  
│   │   ├── feature_engineering.py  
│   │   └── feature_selection.py  
│   ├── models/  
│   │   ├── models_training.py  
│   │   ├── models_testing.py  
│   │   └── models_analysis.py  
│   └── final_results.py  
├── results/  
│   ├── models/  
│   ├── outputs/  
│   ├── plots/  
│   └── selected_features.csv  
└── Reports/  
    └── Project Report
```

Figure 10 - Git : <https://github.com/Fayyaz6137/OECD-WellBeing>

7.2 Feature Selection Logs

Disposable Income (USD PPP)

Social Support (%)

Voter Turnout (%)

Education (%)

Rooms per Person

Employment Rate (%)

Internet Speed Deviation (%)

7.3 Model Hyperparameters

Model	Hyperparameter	Value
Random Forest	n_estimators	100
Random Forest	max_depth	5
Random Forest	random_state	42
Ridge Regression	alpha (baseline)	1.0
Ridge Regression	alpha (tuned)	NA
Ridge Regression	CV folds	5
Gradient Boosting	n_estimators	100
Gradient Boosting	learning_rate	0.1
Gradient Boosting	max_depth	3
Gradient Boosting	random_state	42

Table 3 - Final hyperparameter values used for all trained models.